VIDYAPEETH

BATCH CODE: 12-AN181MA 2026

SUBJECT NAME: PHYSICS

CHAPTER NAME:
UNITS AND MEASURMENT

Lecture No. 9

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Today's Goal

Subtopic Relation b/w two PQ

Subtopic

Subtopic

APPLICATION OF DIMENSIONAL ANALYSIS

1. Principle of HOMOGENITY

2. To convert a physical quantity from one system of measurement to another system of measurement.

3. Deducing relationship between physical quantities.

Q:- Step ① $F \propto m^a \cdot v^b \cdot r^c$

$$\boxed{F = \cancel{K}^{1} \cdot m^a v^b r^c}$$

Step ② write the dimension on Both sides.

$$MLT^{-2} = [M]^a [LT^{-1}]^b [L]^c$$

$$MLT^{-2} = M^a L^b T^{-b} L^c = M^a L^{b+c} T^{-b}$$

$$M^1 L^1 T^{-2} = M^a L^{b+c} T^{-b}$$

Step 3 Equate the powers

$\boxed{a = 1}$; $b + c = 1$; $-b = -2$

$\Rightarrow \boxed{b = 2}$

$2 + c = 1$

$\boxed{c = -1}$

Step ④ Put the values of powers in main Eq^n given in Step ①

$$\boxed{F = m^1 v^2 r^{-1}} \Rightarrow \boxed{F = \frac{mv^2}{r}}$$

Q Force (F), vel (V), & time (T)

$E = F^1 V^1 T^1$

Find [E]

a) $[E] = [F V^2 T]$

b) $[E] = [F V T^2]$

c) $[E] = [F V T]$

d) $[E] = [F V^{-2} T]$

$$\boxed{E \propto F^a V^b T^c}$$

$$ML^2T^{-2} = [MLT^{-2}]^a [LT^{-1}]^b [T]^c$$

$$= M^a L^a T^{-2a} L^b T^{-b} T^c$$

$$M^1 L^2 T^{-2} = M^a L^{a+b} T^{-2a-b+c}$$

Equating powers :-

i) $\boxed{a = 1}$

ii) $a + b = 2$

$\Rightarrow 1 + b = 2$

$\boxed{b = 1}$

(iii) $-2a - b + c = -2$

$-2 - 1 + c = -2$

$c - 3 = -2$

$c = 3 - 2 = 1$

$\boxed{c = 1}$

Q3 E, V

EV^{-2}

[mass] = ?

Q.4 If Energy (E), density (d) & Power (P) are fundamental PQs then find Dimensional formula of Univ. Grav. Const. (G)

$E^{-1} d^{-2} P^{2}$

$E^{2} d^{-1} P^{-1}$

$E^{-2} d^{-1} P^{2}$ ✓

$E^{1} d^{-2} P^{-2}$

Q.5 $U = \dfrac{A\sqrt{x}}{x^2 + B}$; $U \rightarrow$ Pot. Energy
$x \rightarrow$ displacement.

Find [AB] =

$M^{1} L^{11/2} T^{-2}$

SIGNIFICANT FIGURES

SF = 4

$l = 185.2$ m

Certain

Uncertain

185.3

185.001

1.4T : 1000

7.1

7.1
MS

7.100
SG

7.100 Accuracy

7.2

Precision measurment

7.100 cm

7.1 mm

7.12 mm

7.123 mm

7 cm

7.1 cm

7.155

7.2 cm

i) Meter Scale = 0.1 cm
or
1 mm

ii) Vernier Caliper
0.01 cm
0.1 mm

iii) Screw Gauge = 0.01 mm

0.099

SG → 17.11 mm
⇒ 1.711 cm

Topic: Homework-1

- DPP

DPP NO	Question No
DPP -1	
DPP -2	

- Module
  - Exercise Name

Module Page No	Question No
Page No- 21	
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★★★★★

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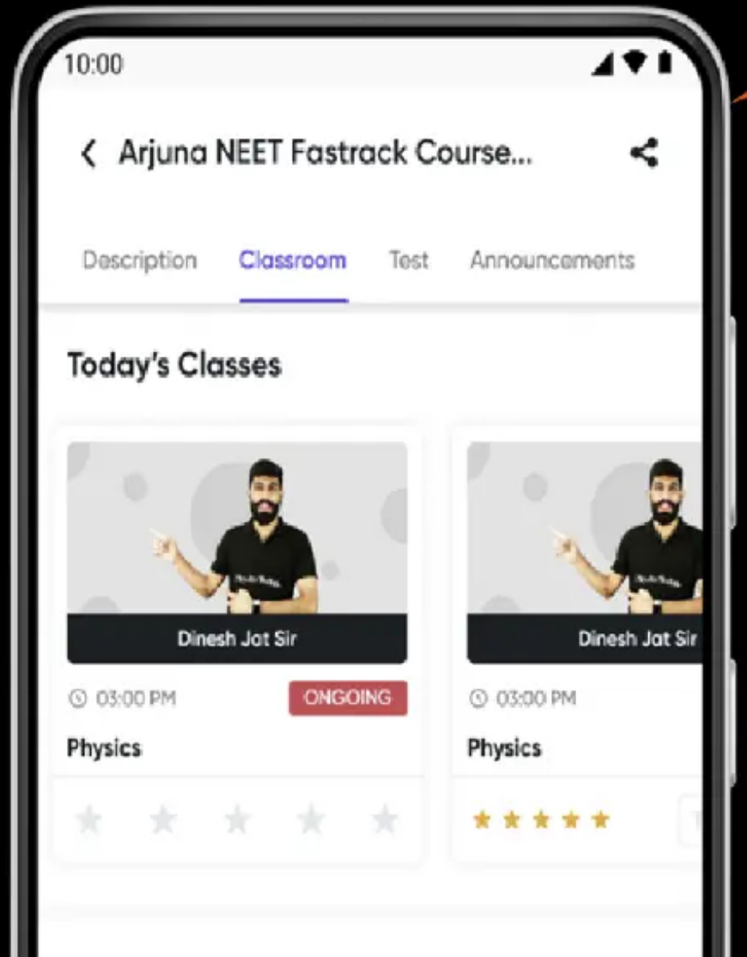
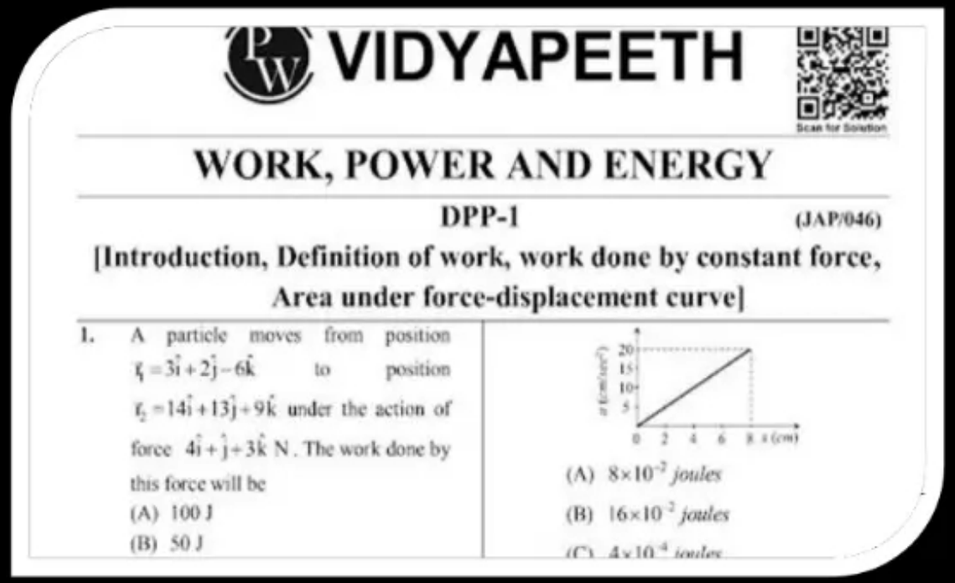


"SCAN" to join our "TELEGRAM" channel

Solve the DPP and check Solution



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Scan for Solution

WORK, POWER AND ENERGY

DPP-1 (JAP/046)

[Introduction, Definition of work, work done by constant force, Area under force-displacement curve]

1. A particle moves from position $\vec{r}_1 = 3\hat{i} + 2\hat{j} - 6\hat{k}$ to position $\vec{r}_2 = 14\hat{i} + 13\hat{j} + 9\hat{k}$ under the action of force $4\hat{i} + \hat{j} + 3\hat{k}$ N. The work done by this force will be

(A) 100 J

(B) 50 J

(A) 8×10^{-2} *joules*

(B) 16×10^{-2} *joules*

Thank You!!!!

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